

# Jordan Mark Barbone, M.A. | Data Scientist, R Developer

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*R programmer with experience in statistical  
programming, monitoring, and report automation in  
cognitive and CNS clinical trials*

*For an updated version see jmbarbone/barbone-cv*

## Experience

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### Data Scientist

**Jun 2021 - Present**

*Cogstate*

*New Haven, CT (Remote)*

- Independently developed internal **R packages** to advance programming, visualization, normative data management, central monitoring, data access, and general utilities
- Designed and development **R shiny** applications for displaying central monitoring activity and performing interactive analyses
- Refactored Data Engineering **PySpark** code into a **python** package for data transformations within **Azure Databricks** and **DeltaTable** management
- Refactored legacy **SQL** code to seamlessly integrate **PySpark** functionalities and enforce best coding practices
- Designed and implemented an efficient process for managing and updating and tracking normative data, collaborating closely with Science and Product Management teams
- Authored comprehensive standards and protocols for **Jira** workflows and **git** management, ensuring consistency and efficiency
- Developed and managed a robust daily and weekly reporting system with **R tidyverse** scripts for the monitoring of a high-profile international Alzheimer's Disease trial
- Analyzed and interpreted requirements from Science and Data Management to drive strategic advancements within our Data Lakehouse
- Developed, reported, and presented analyses with **R tidyverse** scripts for custom data queries related to both ongoing and concluded clinical, providing actionable insights for rater training and central monitoring
- Developed a **python** data extraction, reformatting, and storage processes for academic research studies, greatly enhancing accessibility for end-users
- Contributed to the creation of poster materials for industry conferences and presentations, showcasing research, developments, and services

### Principal Data Analyst

**Dec 2019 - Jun 2021**

*Madrigal Pharmaceuticals*

*Conshohocken, PA*

- Developed internal **R packages** with **tidyverse** features for the creation of daily metric and reports
- Aggregated and organized data sources in STDm, ADaM, TFLS, and pre-finalized formats; compiled source data from multiple vendor data bases and reports; reconciled differences between databases using **R tidyverse** suite
- Analyzed efficacy, safety, and exploratory data for: ongoing phase 3 studies in NASH and NAFLD; ad hoc analyses for phase 2; biomarker and patient reported outcomes with **R**
- Reviewed and reported on statistical testing in **R** such as: ANCOVA, LS-Means, ROC, Kruskal-Wallis, Wilcoxon rank sum, t-test, correlations (Pearson product, Spearman rho, Kendall's tau)
- Trained and onboarded data analyst

### **Clinical Data Scientist**

**Jan 2019 - Nov 2019**

*Signant Health (formerly Bracket and CRF Health)*

*Wayne, PA*

- Developed analytic **R** scripts using **tidyverse** features for clinical trial operational and data management monitoring.
- Created **R** scripts for processing clinical trial outcome data for risk-based statistical quality monitoring
- Provided analytic support for internal and external research projects, including conference poster preparation

### **Clinical Data Specialist**

**Jan 2018 - Dec 2018**

*Signant Health (Bracket)*

*Wayne, PA*

- Analyzed central monitoring data for clinical trials using **R** scripts
- Provided analytic support for internal and external research projects, including conference poster preparation
- Reviewed audio data from ADAS-Cog, MMSE, and ADCS-ADL administrations; provided feedback on standard administration and scoring, and submitted score change reviews for clinical trial sites

## **Education**

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### **Master of Arts**

**West Chester, PA**

*West Chester University of Pennsylvania*

*Class of 2019*

- General Psychology, Advisor: Geeta Shivde, Ph.D.
- Thesis: The effects of participant-selected background music on executive function task performance
- GPA: 4.00

### **Bachelor of Arts**

**West Chester, PA**

*West Chester University of Pennsylvania*

*Class of 2015*

- Major in Psychology, Minor in Biology
- Cumulative GPA: 3.470
- Major GPA: 3.834